

June 5, 2014

Exercise 0. Let $R_1, R_2 \subseteq \Sigma^*$ be regular sets. Show that there is a finite monoid $(M, \circ_M, 1_M)$, monoid homomorphism $h: \Sigma^* \rightarrow M$, and subsets F_1, F_2 of M such that $F_1 \cap F_2 = \emptyset$ and

$$\begin{aligned} R_1 - R_2 &= h^{-1}(F_1), \\ R_2 - R_1 &= h^{-1}(F_2). \end{aligned}$$

Exercise 1. Let $R_1, \dots, R_n \subseteq \Sigma^*$ be regular sets that are pairwise disjoint (i.e., $R_i \cap R_j = \emptyset$ if $i \neq j$). Show that there exists a finite monoid $(M, \circ_M, 1_M)$, a monoid homomorphism $h: \Sigma^* \rightarrow M$, and pairwise disjoint subsets $F_1, \dots, F_n$ of M such that for each $i = 1, \dots, n$, $R_i = h^{-1}(F_i)$.